

34. Hypercomplex Quantities and Representation Theory

Continuation: Chapter I, §§5–7, and Chapter II, §§8–14.

§ 5. Completely reducible groups

A group is called *completely reducible* if it is the direct product of finitely many simple groups:

$$\mathfrak{G} = \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n.$$

In this case the groups

$$\mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n \supset \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_{n-1} \supset \cdots \supset \mathfrak{A}_1 \supset \mathfrak{E}$$

form a composition series. Its length is n , and its composition factors are

$$\sim \mathfrak{A}_n, \mathfrak{A}_{n-1}, \dots, \mathfrak{A}_1.$$

If $\mathfrak{G}$ is completely reducible, then every normal divisor is a direct factor, and the other factor can be chosen as a product of such simple groups as occur in a prescribed product decomposition of $\mathfrak{G}$. The normal divisor $\mathfrak{H}$ is itself completely reducible.

Proof. We have

$$\mathfrak{G} = \mathfrak{H} \mathfrak{A}_1 \mathfrak{A}_2 \cdots \mathfrak{A}_n. \quad (3)$$

Put $\mathfrak{H}_i = \mathfrak{H} \mathfrak{A}_1 \cdots \mathfrak{A}_i$. Then $\mathfrak{H}_{i+1} = \mathfrak{H}_i \mathfrak{A}_{i+1}$. Either $\mathfrak{A}_{i+1} \leq \mathfrak{H}_i$, whence $\mathfrak{H}_{i+1} = \mathfrak{H}_i$; in that case we omit the factor $\mathfrak{A}_{i+1}$ from (3). Or $\mathfrak{H}_i \cap \mathfrak{A}_{i+1}$ is a proper normal divisor of $\mathfrak{A}_{i+1}$, hence is $\mathfrak{E}$, and then $\mathfrak{H}_i \times \mathfrak{A}_{i+1}$ is direct. After the superfluous factors have been omitted, (3) becomes

$$\mathfrak{G} = \mathfrak{H} \times \mathfrak{A}_{i_1} \times \cdots \times \mathfrak{A}_{i_r},$$

so that $\mathfrak{H}$ is a direct factor. Further,

$$\mathfrak{H} \sim \mathfrak{G} / (\mathfrak{A}_{i_1} \times \cdots \times \mathfrak{A}_{i_r}) \sim \mathfrak{A}_{j_1} \times \cdots \times \mathfrak{A}_{j_\mu},$$

where $\mathfrak{A}_{j_1}, \dots, \mathfrak{A}_{j_\mu}$ are the remaining $\mathfrak{A}_i$. Thus $\mathfrak{H}$ is completely reducible.

Corollary. Every decomposition of a completely reducible group into directly indecomposable factors is a decomposition into simple factors, and therefore gives rise to a composition series; from this follows the unique determination of the factors up to isomorphism.

§ 6. Modules with respect to a field. Hypercomplex systems

An almost trivial example of completely reducible groups with operators is given by modules with finite basis with respect to a not necessarily commutative field.

Let $\mathfrak{G}$ be a right K -module, and let the identity element e of K be at the same time the identity operator: $ae = a$ for a in $\mathfrak{G}$. Every submodule aK derived from an a is simple, namely equal to the module derived from any one of its non-zero elements. Thus if $\mathfrak{H}$ is the submodule derived from some elements, and a is an element not belonging to $\mathfrak{H}$, then $\mathfrak{H} \cap aK = \mathfrak{E}$, so $\mathfrak{H} + aK$ is direct. Starting with any basis element a_1 , one can therefore keep adjoining new basis elements $a_2, a_3, \dots$, and obtains $\mathfrak{G}$ as the direct sum

$$\mathfrak{G} = a_1 K + a_2 K + \cdots + a_n K.$$

Thus $\mathfrak{G}$ is completely reducible.

The number n , the length of the composition series, is called the *rank* of $\mathfrak{G}$ with respect to K . Because of the uniqueness of the representation of the elements of $\mathfrak{G}$ (and because e has been assumed to be the identity operator), the basis $a_1, \dots, a_n$ is linearly independent.

Every submodule $\mathfrak{H}$ is a direct summand:

$$\mathfrak{G} = \mathfrak{H} + \mathfrak{R} = (h_1 K + \cdots + h_s K) + (r_1 K + \cdots + r_{n-s} K),$$

or: *every linearly independent basis of $\mathfrak{H}$ can be completed to a linearly independent basis of $\mathfrak{G}$. The r_i may even be chosen from the original basis elements a_i .*

Let $(a_1, \dots, a_n)$ and $(c_1, \dots, c_n)$ be linearly independent bases. Then

$$c_k = \sum_i a_i \pi_{ik},$$

or, written in matrices,

$$(c_1, \dots, c_n) = (a_1, \dots, a_n)P, \quad P = \begin{pmatrix} \pi_{11} & \cdots & \pi_{1n} \\ \vdots & & \vdots \\ \pi_{n1} & \cdots & \pi_{nn} \end{pmatrix}.$$

Conversely,

$$(a_1, \dots, a_n) = (c_1, \dots, c_n)Q.$$

It follows that

$$(a_1, \dots, a_n) = (a_1, \dots, a_n)PQ, \quad (c_1, \dots, c_n) = (c_1, \dots, c_n)QP,$$

and, since the a_i and c_i are linearly independent,

$$PQ = QP = E.$$

Special K -modules that are also rings are the “hypercomplex systems.”

A ring $\mathfrak{o}$ which is at the same time a right module with respect to a commutative field K is called a *hypercomplex system with respect to K* (also called an “algebra over K ” in the literature) if:

1. its rank is finite (let a linearly independent basis be $u_1, \dots, u_n$);
2. $ab \cdot x = a \cdot bx = ax \cdot b$. This is expressed by saying that K is joined commutatively with $\mathfrak{o}$;
3. the identity element e of K is also the identity operator: $ae = a$ for a in $\mathfrak{o}$.

Since

$$\left(\sum_i u_i \alpha_i\right) \left(\sum_k u_k \beta_k\right) = \sum_{i,k} (u_i u_k) (\alpha_i \beta_k),$$

the system is uniquely determined once, in addition to K , the *multiplication table* is given, i.e. once it is known how each product $u_i u_k$ is expressed through the u_j :

$$u_i u_k = \sum_j u_j \gamma_{ik}^j.$$

The γ_{ik}^j must satisfy the familiar relations following from the associative law.

If $\mathfrak{o}$, as we shall often assume, has an identity element e ($ea = ae = a$ for all a), then the elements x of K may be identified with ex , and K may be regarded as a subfield of $\mathfrak{o}$. The hypercomplex system may then also be described as a *ring of finite rank with respect to a field lying in the center*.

An example of a hypercomplex system is the *group ring* of a finite group: one takes the group elements as the basis elements u_i , and group multiplication as multiplication. The field K is arbitrary.

§ 7. Matrices

The square matrices (π_{ik}) , with π_{ik} from a ring $\mathfrak{o}$, form a ring under the ordinary matrix multiplication $\sum_j \pi_{ij} \rho_{jk} = \sigma_{ik}$ and addition $\pi_{ik} + \rho_{ik} = \sigma_{ik}$.

A matrix with elements from a field K for which there is both a right inverse and a left inverse is called *regular*.

We saw in §6: *The transition matrix between two linearly independent bases of a K -module is regular.*

For regularity a right inverse suffices. Indeed, form a right K -module with linearly independent basis $(a_1, \dots, a_n)$ (the module of linear forms in the indeterminates $a_1, \dots, a_n$), and put

$$(c_1, \dots, c_n) = (a_1, \dots, a_n)P.$$

⁰Condition 2 says that multiplication by an element of K is a homomorphism for $\mathfrak{o}$ when $\mathfrak{o}$ is regarded both as an $\mathfrak{o}$ -left-module and as an $\mathfrak{o}$ -right-module. By virtue of 3. these homomorphisms are even isomorphisms.

Then

$$(c_1, \dots, c_n)P^{-1} = (a_1, \dots, a_n)PP^{-1} = (a_1, \dots, a_n),$$

so $(c_1, \dots, c_n)$ is again a basis, hence again linearly independent, and the matrix P is regular. Moreover the right inverse is equal to the left inverse.

Similarly, a left inverse suffices.

If P has a left inverse, then P is not a left zero divisor. Proof. From $PA = 0$ follows $A = P^{-1}PA = 0$.

Thus P also has only one right inverse, which by §6 coincides with the left inverse.

If, as usual, P_x denotes the matrix $(\pi_{ik}x)$, and C_{ik} denotes the matrix which has the identity element in the ik place and zeros everywhere else, then

$$(\pi_{ik}) = \sum_{i,k} C_{ik}\pi_{ik};$$

so the matrices over K form a K -module of rank n^2 . The C_{ik} satisfy the relations

$$C_{ij}C_{jk} = C_{ik}, \quad C_{ij}C_{\bar{j}k} = 0, \quad j \neq \bar{j}.$$

Thus the matrices over K form a finite K -module and, if K is commutative, a hypercomplex system.

Chapter II. Non-commutative ideal theory

§ 8. Homomorphism theorem for rings

If $\mathfrak{a}$ is a two-sided ideal in $\mathfrak{o}$, then the residue-class domain $\mathfrak{o}/\mathfrak{a}$ is not only an $\mathfrak{o}$ -module but also a ring, as is easily seen. Every homomorphic image of $\mathfrak{o}$ is again a ring, and is isomorphic to a residue-class ring $\mathfrak{o}/\mathfrak{a}$, where $\mathfrak{a}$ is a two-sided ideal. The proof is as for groups.

In particular, if $\mathfrak{o}$ is a field, then there are no ideals other than (0) and $\mathfrak{o}$; hence here the homomorphism is either an isomorphism, or assigns zero to every element.

In particular, suppose the ring $\mathfrak{o}$ is a right K -module, where K is to be a ring elementwise commuting with $\mathfrak{o}$:

$$ab \cdot x = a \cdot bx = ax \cdot b$$

(for example, in the case of a hypercomplex system with respect to a commutative field K). Then as admissible right, left, or two-sided ideals one considers only those which are at the same time K -modules, and in addition to ring homomorphism one also requires operator homomorphism with respect to multiplication by K . The admissible ideals become bimodules with respect to $\mathfrak{o}$ and K . (For hypercomplex systems, by “ideals” without further qualification one always means only those admissible with respect to the underlying coefficient field K .) In this enlarged sense too, the homomorphism theorem above remains valid.

An example of ring homomorphism is the transition from a multiplier domain $\mathfrak{o}$ to the absolute multiplier domain (§1, Example 4). Thus the absolute multiplier domain is isomorphic to the residue-class ring $\mathfrak{o}/\mathfrak{d}$, where $\mathfrak{d}$ is the two-sided ideal of elements of $\mathfrak{o}$ which annihilate $\mathfrak{M}$.

From the validity of the homomorphism theorem there follow, as in §2, the first and second isomorphism theorems for ring and operator isomorphism.

All products $\mathfrak{A}\mathfrak{B}$ of subsets of $\mathfrak{o}$ are to be understood as module products: $\mathfrak{A}\mathfrak{B}$ is the totality of all sums $\sum ab$, with a in $\mathfrak{A}$, b in $\mathfrak{B}$, and $a\mathfrak{B}$ is the totality of all sums $\sum ab$, b in $\mathfrak{B}$. If $\mathfrak{B}$ is a right module with respect to some ring as operator domain, then the product $\mathfrak{A}\mathfrak{B}$, respectively $a\mathfrak{B}$, is also a right module. In particular: a right ideal, an admissible ideal with respect to a coefficient domain K , etc. If $(\mathfrak{A}, \mathfrak{B})$ denotes the sum of the modules $\mathfrak{A}$ and $\mathfrak{B}$, then the rules of calculation are

$$\mathfrak{A}\mathfrak{B} \cdot \mathfrak{C} = \mathfrak{A} \cdot \mathfrak{B}\mathfrak{C},$$

$$\mathfrak{A}(\mathfrak{B}, \mathfrak{C}) = (\mathfrak{A}\mathfrak{B}, \mathfrak{A}\mathfrak{C}), \quad (\mathfrak{A}, \mathfrak{B})\mathfrak{C} = (\mathfrak{A}\mathfrak{C}, \mathfrak{B}\mathfrak{C}).$$

The direct sum is denoted by $\mathfrak{A} + \mathfrak{B}$ (§4).

In what follows we shall often consider rings satisfying a maximal or minimal condition (§3) for right ideals. Hypercomplex systems in particular belong to these, since by the convention just made only K -modules are admitted as ideals, and their rank therefore remains bounded (§6).

§ 9. Idempotent elements. Direct sum decomposition into right ideals

Let $\mathfrak{o}$ be a ring with identity element.

If $\mathfrak{r}$ is a right ideal, then $\mathfrak{r}\mathfrak{o} \subseteq \mathfrak{r}$, and because of the identity element even $\mathfrak{r}\mathfrak{o} = \mathfrak{r}$. Correspondingly for left ideals: $\mathfrak{o}\mathfrak{l} = \mathfrak{l}$.

An element c is called idempotent if $c^2 = c$.

If $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$ is a decomposition into right ideals, and if

$$e = e_1 + \cdots + e_n, \quad e_i \in \mathfrak{r}_i,$$

then

$$e_i^2 = e_i, \quad e_i e_k = 0 \quad (i \neq k) \quad (\text{"orthogonality relations"}),$$

$$\mathfrak{r}_i = e_i \mathfrak{o}.$$

Proof. Let r be an element of $\mathfrak{r}_1$. Then

$$r = er = e_1 r + \cdots + e_n r, \quad e_i r \in \mathfrak{r}_i,$$

but also

$$r = r + 0 + \cdots + 0,$$

so

$$e_1 r = r, \quad e_i r = 0 \quad (i \neq 1).$$

From the first formula it follows that $\mathfrak{r}_1 \subseteq e_1 \mathfrak{o}$; on the other hand $e_1 \mathfrak{o} \subseteq \mathfrak{r}_1$, so $e_1 \mathfrak{o} = \mathfrak{r}_1$. Specializing $r = e_1$ gives

$$e_1^2 = e_1, \quad e_i e_1 = 0 \quad (i \neq 1).$$

The same holds when the index 1 is replaced by k .

Conversely, if $e = \sum e_i$, $e_i e_k = 0$ ($i \neq k$), $e_i^2 = e_i$, and one puts $\mathfrak{r}_i = e_i \mathfrak{o}$, then $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$.

Proof. Every element r of $\mathfrak{o}$ is

$$r = er = e_1 r + \cdots + e_n r,$$

hence

$$\mathfrak{o} = (e_1 \mathfrak{o}, \dots, e_n \mathfrak{o}).$$

But the representation is unique; for from

$$0 = e_1 a_1 + \cdots + e_n a_n$$

it follows, upon multiplication by e_i , that

$$e_i^2 a_i = e_i a_i = 0.$$

From a right representation

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = e_1 \mathfrak{o} + \cdots + e_n \mathfrak{o}$$

there accordingly follows a left representation

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n = \mathfrak{o} e_1 + \cdots + \mathfrak{o} e_n.$$

If the $\mathfrak{r}_i$ are directly indecomposable, then so are the $\mathfrak{l}_i$. For a decomposition, say of $\mathfrak{l}_1$, would mean

$$\mathfrak{l}_1 = \mathfrak{o} e_1 = \mathfrak{o} e'_1 + \mathfrak{o} e''_1, \quad \mathfrak{o} = \mathfrak{o} e'_1 + \mathfrak{o} e''_1 + \mathfrak{o} e_2 + \cdots + \mathfrak{o} e_n.$$

From this would result the right decomposition

$$\mathfrak{o} = e'_1 \mathfrak{o} + e''_1 \mathfrak{o} + e_2 \mathfrak{o} + \cdots + e_n \mathfrak{o} = \mathfrak{r}'_1 + \mathfrak{r}''_1 + \mathfrak{r}_2 + \cdots + \mathfrak{r}_n,$$

$$\mathfrak{r}_1 \cong \mathfrak{o} / (\mathfrak{r}_2 + \cdots + \mathfrak{r}_n) = \mathfrak{r}'_1 + \mathfrak{r}''_1,$$

so $\mathfrak{r}_1$ would be decomposable.

⁰There is another converse: if $e_i e_k = 0$, $e_i^2 = e_i$, $c = \sum e_i$, then c is a left identity for the ring $e_1 \mathfrak{o} + \cdots + e_n \mathfrak{o}$. The fact that the sum is direct is seen in exactly the same way as above.

§ 10. Decomposition into two-sided ideals

Let $\mathfrak{o}$ again be a ring with identity element, and let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n \quad (1)$$

be a decomposition of $\mathfrak{o}$ into two-sided ideals. Then the orthogonality relations also hold for the ideals:

$$\mathfrak{a}_i \mathfrak{a}_k = 0 \quad (i \neq k), \quad \mathfrak{a}_i^2 = \mathfrak{a}_i. \quad (2)$$

Proof. Let $\mathfrak{b}_1 = \mathfrak{a}_2 + \cdots + \mathfrak{a}_n$, $\mathfrak{o} = \mathfrak{a}_1 + \mathfrak{b}_1$. Then $\mathfrak{a}_1 \mathfrak{b}_1 \leq \mathfrak{a}_1 \cap \mathfrak{b}_1 = 0$, so $\mathfrak{a}_1 \mathfrak{b}_1 = 0$, and a fortiori

$$\mathfrak{a}_1 \mathfrak{a}_i = 0 \quad (i \neq 1);$$

$$\mathfrak{a}_1 = \mathfrak{a}_1 \mathfrak{o} = \mathfrak{a}_1 (\mathfrak{a}_1 + \mathfrak{b}_1) = (\mathfrak{a}_1^2, \mathfrak{a}_1 \mathfrak{b}_1) = \mathfrak{a}_1^2.$$

Conversely: from (1) and (2) it follows that the modules $\mathfrak{a}_i$ are two-sided ideals. *Proof.*

$$\mathfrak{o} \mathfrak{a}_i = (\mathfrak{a}_1 + \cdots + \mathfrak{a}_n) \mathfrak{a}_i = \mathfrak{a}_i^2 = \mathfrak{a}_i, \quad \mathfrak{a}_i \mathfrak{o} = \mathfrak{a}_i.$$

Right ideals in the ring $\mathfrak{a}_i$ are right ideals in $\mathfrak{o}$. Proof.

$$\mathfrak{r} \mathfrak{o} = \mathfrak{r} (\mathfrak{a}_i + \mathfrak{b}_i) = (\mathfrak{r} \mathfrak{a}_i, \mathfrak{r} \mathfrak{b}_i) = (\mathfrak{r}, 0) = \mathfrak{r}.$$

If $\mathfrak{r}$ is a right ideal in $\mathfrak{o}$, then $\mathfrak{r}$ is a direct sum of right ideals $\mathfrak{r} \mathfrak{a}_i \leq \mathfrak{a}_i$. Proof. $\mathfrak{r} = \mathfrak{r} \mathfrak{o} = (\mathfrak{r} \mathfrak{a}_1, \dots, \mathfrak{r} \mathfrak{a}_n)$, and $\mathfrak{r} \mathfrak{a}_i \mathfrak{o} = \mathfrak{r} \mathfrak{a}_i$, so the $\mathfrak{r} \mathfrak{a}_i$ are right ideals contained in $\mathfrak{r}$. Since $\mathfrak{r} \mathfrak{a}_i \leq \mathfrak{a}_i$, the sum is direct:

$$\mathfrak{r} = \mathfrak{r} \mathfrak{a}_1 + \cdots + \mathfrak{r} \mathfrak{a}_n.$$

In particular, if $\mathfrak{r}$ is directly indecomposable, then $\mathfrak{r}$ can have only one component; hence $\mathfrak{r}$ must lie in one of the $\mathfrak{a}_i$.

By means of these theorems, ideal theory in $\mathfrak{o}$ is controlled once the ideal theory of the individual $\mathfrak{a}_i$ is known. We shall therefore mostly restrict ourselves to two-sided indecomposable rings.

Two right ideals contained in different $\mathfrak{a}_i$ can never be operator-isomorphic. Proof. An ideal lying in $\mathfrak{a}_1$ is annihilated by all the other $\mathfrak{a}_k$, but not by $\mathfrak{a}_1$. By contrast, an ideal lying in $\mathfrak{a}_2$ is annihilated by $\mathfrak{a}_1$.

Thus if it is possible to decompose the ring $\mathfrak{o}$ into two-sided indecomposable ideals $\mathfrak{a}_1 + \cdots + \mathfrak{a}_n$, and each $\mathfrak{a}_i$ again into one-sided indecomposable right ideals $\mathfrak{r}'_i + \mathfrak{r}''_i + \cdots$, then the operator-isomorphic $\mathfrak{r}$ are to be sought among those belonging to the same $\mathfrak{a}_i$. But it may happen that there are still different classes of $\mathfrak{r}$ – that is, classes not operator-isomorphic – in the same $\mathfrak{a}_i$, as the following example shows.

Let K be the field of rational numbers,

$$\mathfrak{o} = e_1 K + e_2 K + u K$$

a hypercomplex system, with the e_i and u commuting with the numbers of K , and with multiplication table

	e_1	e_2	u
e_1	e_1	0	u
e_2	0	e_2	0
u	0	u	0

The identity element is $e = e_1 + e_2$. The e_i satisfy the orthogonality relations. Thus the right decomposition is

$$\mathfrak{o} = e_1 \mathfrak{o} + e_2 \mathfrak{o} = (e_1, u) + (e_2),$$

and the left decomposition is

$$\mathfrak{o} = \mathfrak{o} e_1 + \mathfrak{o} e_2 = (e_1) + (e_2, u).$$

Since $e_2 \mathfrak{o}$ and $\mathfrak{o} e_1$ are visibly indecomposable, $\mathfrak{o} e_2$ and $e_1 \mathfrak{o}$ must be indecomposable as well.

A two-sided decomposition is impossible. For then one component would have to contain $e_1 \mathfrak{o}$ and the other $e_2 \mathfrak{o}$; the first would then contain u , but the second would contain $u e_2 = u$ as well.

But $e_1 \mathfrak{o}$ and $e_2 \mathfrak{o}$ are not operator-isomorphic, since they have different rank with respect to K .

Let $\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n$, with the $\mathfrak{a}_i$ two-sided indecomposable. Then they are uniquely determined.

Proof. Suppose also $\mathfrak{o} = \mathfrak{c}_1 + \cdots + \mathfrak{c}_n$. Then

$$\mathfrak{a}_i = \mathfrak{a}_i \mathfrak{o} = (\mathfrak{a}_i \mathfrak{c}_1, \dots, \mathfrak{a}_i \mathfrak{c}_n).$$

This latter sum is direct because $\mathfrak{a}_i \mathfrak{c}_k \leq \mathfrak{c}_k$ and $\leq \mathfrak{a}_i$; since, however, the $\mathfrak{a}_i$ are indecomposable, all $\mathfrak{a}_i \mathfrak{c}_k$ must be zero except one, $\mathfrak{a}_i \mathfrak{c}_{j_i}$. Thus

$$\mathfrak{a}_i = \mathfrak{a}_i \mathfrak{c}_{j_i}.$$

Similarly,

$$\mathfrak{c}_{j_i} = \mathfrak{o} \mathfrak{c}_{j_i} = \mathfrak{a}_1 \mathfrak{c}_{j_i} + \cdots + \mathfrak{a}_n \mathfrak{c}_{j_i},$$

and since $\mathfrak{a}_i \mathfrak{c}_{j_i} \neq 0$, all the remaining terms are 0. Therefore

$$\mathfrak{c}_{j_i} = \mathfrak{a}_i \mathfrak{c}_{j_i} = \mathfrak{a}_i,$$

q.e.d.

§ 11. The center

The center $\mathfrak{Z}$ of a ring $\mathfrak{o}$ is the totality of all z commuting with all ring elements ($za = az$ for all a). $\mathfrak{Z}$ is a commutative ring.

Every one-sided ideal $\mathfrak{a}$ in $\mathfrak{o}$ has a “contraction ideal” $\mathfrak{a} \cap \mathfrak{Z}$ in $\mathfrak{Z}$. Since both $\mathfrak{a}$ and $\mathfrak{Z}$ are $\mathfrak{Z}$ -modules, so is $\mathfrak{a} \cap \mathfrak{Z}$.

Every ideal $\mathfrak{A}$ in $\mathfrak{Z}$ has an extension ideal: the ideal $\mathfrak{a} = (\mathfrak{A}, \mathfrak{o}\mathfrak{A})$ generated by $\mathfrak{A}$ in $\mathfrak{o}$. This is two-sided, because

$$\mathfrak{a}\mathfrak{o} = (\mathfrak{o}\mathfrak{A}, \mathfrak{A})\mathfrak{o} = (\mathfrak{o}\mathfrak{A}\mathfrak{o}, \mathfrak{A}\mathfrak{o}) = (\mathfrak{o}^2\mathfrak{A}, \mathfrak{o}\mathfrak{A}) \subseteq \mathfrak{o}\mathfrak{A} \subseteq \mathfrak{a}.$$

If $\mathfrak{o}$ is assumed to have an identity element, one may write $\mathfrak{a} = \mathfrak{o}\mathfrak{A}$. In this case the following theorem holds:

Every two-sided decomposition of $\mathfrak{o}$ corresponds one-to-one to a decomposition of the center: from

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n, \quad \mathfrak{A}_i = \mathfrak{a}_i \cap \mathfrak{Z} \tag{1}$$

there follows

$$\mathfrak{Z} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_n, \quad \mathfrak{o}\mathfrak{A}_i = \mathfrak{a}_i, \tag{2}$$

and conversely.

Proof. Let z be a central element, and let

$$z = z_1 + \cdots + z_n \tag{3}$$

be its decomposition according to (1). Then the z_i are central elements; for

$$za = z_1 a + \cdots + z_n a = az = az_1 + \cdots + az_n.$$

By directness of the sum, and since $\mathfrak{a}_i$ is two-sided, it follows that

$$z_i a = az_i.$$

Thus z_i lies in $\mathfrak{a}_i \cap \mathfrak{Z} = \mathfrak{A}_i$, so (3) gives the asserted sum decomposition. In particular, $e = \sum e_i$, hence the e_i are central elements. Finally

$$z = ze_1 + \cdots + ze_n,$$

and so

$$\mathfrak{A}_i = \mathfrak{Z}e_i, \quad \mathfrak{o}\mathfrak{A}_i = \mathfrak{o}\mathfrak{Z}e_i = \mathfrak{o}e_i = \mathfrak{a}_i.$$

Conversely, if $\mathfrak{Z} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_n$ is a decomposition of the center, then by §9

$$\mathfrak{A}_i = \mathfrak{Z}e_i, \quad e_i^2 = e_i, \quad e_i e_k = 0 \quad (i \neq k).$$

Hence, by the orthogonality relations §9,

$$\begin{aligned} \mathfrak{o} &= \mathfrak{o}e = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n = \mathfrak{o}\mathfrak{Z}e_1 + \cdots + \mathfrak{o}\mathfrak{Z}e_n \\ &= \mathfrak{o}\mathfrak{A}_1 + \cdots + \mathfrak{o}\mathfrak{A}_n = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n. \end{aligned}$$

If now $\mathfrak{a}_i \cap \mathfrak{Z} = \mathfrak{A}'_i$, then from the first part of the assertion one knows that

$$\mathfrak{Z} = \mathfrak{A}'_1 + \cdots + \mathfrak{A}'_n \quad \text{directly.}$$

But $\mathfrak{A}'_i \supseteq \mathfrak{A}_i$, hence $\mathfrak{A}'_i = \mathfrak{A}_i$ by §4, 2.

Corollary. $\mathfrak{a}_i$ and $\mathfrak{A}_i$ are simultaneously two-sided directly decomposable or indecomposable.

§ 12. Nilpotent ideals

An ideal $\mathfrak{c}$ is called *nilpotent* if $\mathfrak{c}^\rho = 0$.

Example. The ideal (u) in §10.

If there is a nilpotent right ideal $\mathfrak{r}$ in $\mathfrak{o}$, then there is also a nilpotent left ideal, indeed a two-sided ideal.

Proof. Let $\mathfrak{r}^\rho = 0$. Then $\mathfrak{o}\mathfrak{r}$, or, if $\mathfrak{o}$ contains no identity element and $\mathfrak{o}\mathfrak{r}$ could be zero, $(\mathfrak{o}\mathfrak{r}, \mathfrak{r})$, is a left ideal, and

$$(\mathfrak{o}\mathfrak{r})^\rho = \mathfrak{o}\mathfrak{r}\mathfrak{o}\mathfrak{r}\cdots\mathfrak{o}\mathfrak{r} \subseteq \mathfrak{o}\mathfrak{r}\mathfrak{r}\cdots\mathfrak{r} = \mathfrak{o}\mathfrak{r}^\rho = 0.$$

The sum of two nilpotent right ideals is again a nilpotent right ideal.

Proof. Let $\mathfrak{c}^\rho = 0$, $\mathfrak{d}^\sigma = 0$. In

$$(\mathfrak{c}, \mathfrak{d})^{\rho+\sigma-1} = (\dots, \mathfrak{d}\mathfrak{c}\cdots\mathfrak{d}\cdots\mathfrak{c}, \dots)$$

every term contains either at least ρ factors $\mathfrak{c}$ or at least σ factors $\mathfrak{d}$. In the first case one has

$$\mathfrak{d}\mathfrak{c}\cdots\mathfrak{d}\cdots\mathfrak{c}\cdots \subseteq \mathfrak{d}\mathfrak{c}\mathfrak{c}\cdots\mathfrak{c} = \mathfrak{d}\mathfrak{c}^\rho = 0;$$

in the latter case the same argument applies to the factors $\mathfrak{d}$. Thus

$$(\mathfrak{c}, \mathfrak{d})^{\rho+\sigma-1} = 0.$$

If now the maximal condition for right ideals in $\mathfrak{o}$ is satisfied, then there is a maximal nilpotent right ideal. It contains all other nilpotent right ideals; otherwise one could form its sum with such an ideal. Call it $\mathfrak{c}$. Since $\mathfrak{o}\mathfrak{c}$ is also a nilpotent right ideal, $\mathfrak{o}\mathfrak{c} \subseteq \mathfrak{c}$, so $\mathfrak{c}$ is a left ideal. Every nilpotent left ideal $\mathfrak{d}$ is likewise contained in $\mathfrak{c}$, since $(\mathfrak{d}, \mathfrak{d}\mathfrak{o})$ is a nilpotent right ideal. Thus $\mathfrak{d} \subseteq \mathfrak{c}$. *There is therefore a maximal two-sided nilpotent ideal $\mathfrak{c}$, or “radical,” containing all the others, right- and left-sided.*

In the example of §10, (u) is the maximal nilpotent ideal.

If the radical is the zero ideal, one speaks of a “ring without radical” (“semisimple ring,” “Dedekind system”). The residue-class ring modulo the radical is always a ring without radical.

If $\mathfrak{Z}$ is the center of $\mathfrak{o}$, and $\mathfrak{c}$ is the radical of $\mathfrak{o}$, then $\mathfrak{C} = \mathfrak{c} \cap \mathfrak{Z}$ is the radical of $\mathfrak{Z}$.

Proof. It is clear that $\mathfrak{C}$ is nilpotent. If there were a larger nilpotent ideal $\bar{\mathfrak{C}}$ in $\mathfrak{Z}$, then it could be extended to a nilpotent ideal $\bar{\mathfrak{c}}$ not wholly contained in $\mathfrak{c}$, since $((\bar{\mathfrak{C}}\mathfrak{o})^\rho = \bar{\mathfrak{C}}^\rho \mathfrak{o}^\rho = 0)$.

Consequence. If $\mathfrak{o}$ is a ring without radical, so is $\mathfrak{Z}$. The converse is not true, however, as the example in §10 shows. The center of $\mathfrak{o}/\mathfrak{c}$ contains a ring isomorphic to $\mathfrak{Z}/\mathfrak{C}$, but this can be a proper subring (example in §10).

§ 13. Completely reducible rings

By §5 a ring is called *right completely reducible* if it is a direct sum of finitely many simple right ideals.

A right completely reducible ring with identity element has no nilpotent ideal $\neq (0)$, hence no radical.

Proof. Every right ideal $\mathfrak{r}$ is a direct summand (§5):

$$\mathfrak{o} = \mathfrak{t} + \mathfrak{r} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

Since $e_2^2 = e_2$, also $e_2^\rho = e_2$. If $\mathfrak{r}$ were nilpotent, then $e_2^\rho = 0$, hence $e_2 = 0$, and so $\mathfrak{r} = 0$, q.e.d.

We now show the two converses. First: *A ring without radical satisfying the minimal condition for right ideals is completely reducible with respect to right ideals.*

Proof. Let $\mathfrak{t}$ be a minimal right ideal $\neq 0$. We shall show that $\mathfrak{t}$ contains an idempotent element.

Since $\mathfrak{t}^2 \subseteq \mathfrak{t}$ and $\mathfrak{t}^2 \neq 0$, one has $\mathfrak{t}^2 = \mathfrak{t}$, for $\mathfrak{t}^2$ is a right ideal. Hence there is an a in $\mathfrak{t}$ such that $a\mathfrak{t} \neq 0$. Then necessarily $a\mathfrak{t} = \mathfrak{t}$, because $a\mathfrak{t}$ is a right ideal.

The totality of all b in $\mathfrak{t}$ which are annihilated by a ($ab = 0$) is a right ideal, and is not equal to $\mathfrak{t}$, hence is 0. Thus $ab = 0$ implies $b = 0$.

Because $\mathfrak{t} = a\mathfrak{t}$, a must have a representation $a = ac$, with $c \neq 0$. It follows that

$$ac = ac^2, \quad a(c - c^2) = 0, \quad c - c^2 = 0, \quad c^2 = c.$$

We next show that $\mathfrak{t}$ is a direct summand. The set $c\mathfrak{o}$ is a non-zero right ideal contained in $\mathfrak{t}$, since $c^2 = c$ lies in $c\mathfrak{o}$; hence it is $\mathfrak{t}$. Every element r of $\mathfrak{o}$ has a representation

$$r = cr + (r - cr) \quad (\text{one-sided Peirce decomposition}).$$

The elements cr form the ideal $\mathfrak{t}$; the elements $r - cr$ form another right ideal $\mathfrak{r}$, annihilated by c : $c\mathfrak{r} = 0$. Hence

$$\mathfrak{o} = (\mathfrak{t}, \mathfrak{r}).$$

Since the elements of $\mathfrak{t}$ are not annihilated by c , the sum is direct:

$$\mathfrak{o} = \mathfrak{t} + \mathfrak{r}. \quad (1)$$

Representing $\mathfrak{o}$, by the minimal condition, as a direct sum of directly indecomposable ideals,

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n,$$

the $\mathfrak{r}_i$ must be simple, i.e. minimal. For if, say, $\mathfrak{r}_1$ were not simple and $\mathfrak{t}$ were a minimal ideal in $\mathfrak{r}_1$, then (1) would give

$$\mathfrak{r}_1 = \mathfrak{t} + \mathfrak{r} \cap \mathfrak{r}_1 \quad \S 4, 3,$$

so $\mathfrak{r}_1$ would be decomposable, a contradiction. Thus $\mathfrak{o}$ is completely reducible.

Second: *A ring without radical satisfying the minimal condition has an identity element.*

To show first the existence of a left identity, it is enough to prove the following: *if a right ideal $\mathfrak{a} = c_1\mathfrak{o} \neq \mathfrak{o}$ has a left identity c_1 , then there is a right ideal $c\mathfrak{o}$ of greater length which likewise has a left identity c .* For, since complete reducibility, and therefore boundedness of all lengths, has already been shown, one reaches in finitely many steps a left identity for $\mathfrak{o}$ itself.

Let therefore $\mathfrak{a} = c_1\mathfrak{o}$ and $c_1^2 = c_1$. By the formula

$$r = c_1r + (r - c_1r)$$

we have, as above, a Peirce decomposition $\mathfrak{o} = c_1\mathfrak{o} + \mathfrak{r}$. Let c_2 be an idempotent element of $\mathfrak{r}$, as above; thus $c_2^2 = c_2$, and $c_1c_2 = 0$ since c_1 annihilates all elements of $\mathfrak{r}$. Put $e_1 = c_1$, $e_2 = c_2 - c_2c_1$. Then

$$e_1^2 = e_1, \quad e_1e_2 = 0, \quad e_2e_1 = 0, \quad e_2c_2 = e_2, \quad e_2 \neq 0, \quad e_2^2 = e_2.$$

By the remark made in the note above, $c = e_1 + e_2$ is a left identity for the ring

$$c\mathfrak{o} = e_1\mathfrak{o} + e_2\mathfrak{o},$$

which, since $e_2^2 = e_2 \neq 0$, has greater length than $\mathfrak{a}$ as a right ideal.

To show that the constructed left identity e is also a right identity, we make the Peirce decomposition into left ideals:

$$\mathfrak{o} = \mathfrak{o}e + \mathfrak{l}.$$

We have $\mathfrak{l}e = 0$, $\mathfrak{l} = e\mathfrak{l}$, hence $\mathfrak{l}^2 = \mathfrak{l}e\mathfrak{l} = 0$. Since by hypothesis no nilpotent left ideal can exist, $\mathfrak{l} = 0$. Therefore e is also a right identity, and hence an identity altogether.

Theorem. From right-sided complete reducibility and the existence of the identity follows two-sided complete reducibility:

$$\mathfrak{o} = \mathfrak{d}_1 + \cdots + \mathfrak{d}_s, \quad \mathfrak{d}_i \text{ two-sided simple.}$$

As in the preceding proof, it is enough to show that every minimal two-sided ideal $\mathfrak{a}$ is a direct summand. As a right ideal, $\mathfrak{a}$ is a direct summand:

$$\mathfrak{o} = \mathfrak{a} + \mathfrak{r} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

The corresponding left decomposition is

$$\mathfrak{o} = \mathfrak{c} + \mathfrak{l} = \mathfrak{o}e_1 + \mathfrak{o}e_2.$$

It follows that

$$\begin{aligned} \mathfrak{r}\mathfrak{c} &\subseteq \mathfrak{r}\mathfrak{a} \leq \mathfrak{r} \cap \mathfrak{a} = 0, & \mathfrak{l}\mathfrak{a} &= \mathfrak{o}e_2e_1\mathfrak{o} = 0, \\ (\mathfrak{a}\mathfrak{l})^2 &= \mathfrak{a}\mathfrak{l}\mathfrak{a}\mathfrak{l} = 0, & \text{hence } \mathfrak{a}\mathfrak{l} &= 0; \\ \mathfrak{r} &= \mathfrak{r}\mathfrak{o} = (\mathfrak{r}\mathfrak{c}, \mathfrak{r}\mathfrak{l}) = \mathfrak{r}\mathfrak{l}, & \mathfrak{l} &= \mathfrak{o}\mathfrak{l} = (\mathfrak{a}\mathfrak{l}, \mathfrak{r}\mathfrak{l}) = \mathfrak{r}\mathfrak{l}. \end{aligned}$$

Thus $\mathfrak{r} = \mathfrak{l}$, so $\mathfrak{r}$ is two-sided; hence $\mathfrak{a}$ is a two-sided direct summand. From here on the proof is the same as for one-sided complete reducibility.

The $\mathfrak{d}_i$ are also two-sided simple as rings, since all ideals in $\mathfrak{d}_i$ are ideals in $\mathfrak{o}$. We now investigate their structure.

§ 14. Two-sided simple completely reducible rings with identity element

Let $\mathfrak{o}$ be such a ring:

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = e_1 \mathfrak{o} + \cdots + e_n \mathfrak{o}, \quad \mathfrak{r}_i \text{ simple.}$$

Then

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n, \quad \mathfrak{l}_i \text{ directly indecomposable §9.}$$

Moreover $\mathfrak{l}_i \mathfrak{r}_i$ is a non-zero two-sided ideal (since e_i^2 lies in $\mathfrak{l}_i \mathfrak{r}_i$), hence is $\mathfrak{o}$.

Put $\mathfrak{A}_{ik} = \mathfrak{r}_i \mathfrak{l}_k$. Then

$$\mathfrak{o} = (\mathfrak{r}_1, \dots, \mathfrak{r}_n) \cdot (\mathfrak{l}_1, \dots, \mathfrak{l}_n) = (\dots, \mathfrak{A}_{ij}, \dots)$$

directly. For if $0 = \sum_{i,j} a_{ij}$, then, since a_{ij} lies in $\mathfrak{r}_i$ and $\sum \mathfrak{r}_i$ is direct,

$$0 = \sum_j a_{ij},$$

and then, because a_{ij} lies in $\mathfrak{l}_j$,

$$0 = a_{ij}.$$

Further, $\mathfrak{A}_{ij} \mathfrak{r}_j = \mathfrak{r}_i \mathfrak{l}_j \mathfrak{r}_j = \mathfrak{r}_i \mathfrak{o} = \mathfrak{r}_i$, so $\mathfrak{A}_{ij} \neq 0$. Let $a_{ij} \neq 0$ lie in $\mathfrak{A}_{ij}$. Then

$$a_{ij} \mathfrak{r}_j \subseteq \mathfrak{r}_i, \quad a_{ij} \mathfrak{r}_j \neq 0 \text{ because } a_{ij} e_j = a_{ij},$$

and therefore

$$a_{ij} \mathfrak{r}_j = \mathfrak{r}_i.$$

If for every r_j in $\mathfrak{r}_j$ one puts

$$a_{ij} r_j = r_i,$$

then the map $r_j \mapsto r_i$ is an operator homomorphism from $\mathfrak{r}_j$ to $\mathfrak{r}_i$. The elements mapped to zero form an ideal contained in $\mathfrak{r}_j$, hence the zero ideal; therefore the map is an *isomorphism*. Thus:

Any two simple right ideals $\mathfrak{r}_i$ and $\mathfrak{r}_j$ occurring in a representation of $\mathfrak{o}$ are operator-isomorphic; the isomorphisms are mediated by the elements of $\mathfrak{A}_{ij}$. Since any two different simple right ideals $\mathfrak{r}, \mathfrak{r}'$ occur together in at least one representation of $\mathfrak{o}$, every two such ideals are isomorphic.

All homomorphisms from $\mathfrak{r}_j$ to $\mathfrak{r}_i$ are mediated by elements of $\mathfrak{A}_{ij}$.

Proof. If $e_j \mapsto a_{ij}$, then $e_j^2 \mapsto a_{ij} e_j$, so $a_{ij} = a_{ij} e_j$ lies in $\mathfrak{r}_i \mathfrak{l}_j = \mathfrak{A}_{ij}$. Further,

$$\mathfrak{r}_j = e_j \mathfrak{r}_j \mapsto a_{ij} \mathfrak{r}_j = \mathfrak{r}_i.$$

In particular, the homomorphisms of $\mathfrak{r}_i$ into itself are mediated by $\mathfrak{A}_{ii}$.

The product of two elements of $\mathfrak{A}_{ii}$ corresponds to the product of the homomorphisms, and the sum to the sum (definition: see §1, 6). Distinct a_{ii} give distinct automorphisms, for e_i goes to $a_{ii} e_i = a_{ii}$. Thus the ring $\mathfrak{A}_{ii}$ is isomorphic to the automorphism ring of $\mathfrak{r}_i$.

But the automorphism ring of a simple ideal is a field (§2), hence $\mathfrak{A}_{ii}$ is a field. Since all $\mathfrak{r}_i$ are operator-isomorphic, their automorphism rings are ring-isomorphic: $\mathfrak{A}_{ii}$ is uniquely determined by $\mathfrak{o}$ up to ring isomorphism. The possible different $\mathfrak{A}_{ii}$ are only different concrete realizations of the abstractly defined automorphism ring of the simple right ideals.

Main theorem. Every completely reducible two-sided simple ring $\mathfrak{o}$ with identity element is isomorphic to the ring of matrices of degree n over a field K . (The field K is isomorphic to the automorphism field of the right ideals of $\mathfrak{o}$.)

Proof. Construction of the matrix units c_{ik} . Let Γ_{11} be the identical automorphism of $\mathfrak{r}_1$, let Γ_{i1} be an arbitrary isomorphism $\Gamma_{i1} \mathfrak{r}_1 = \mathfrak{r}_i$, and finally set

$$\Gamma_{ik} = \Gamma_{i1} \Gamma_{k1}^{-1}.$$

Then in general

$$\Gamma_{ik} \Gamma_{kl} = \Gamma_{il}.$$

If Γ_{ik} is mediated by the element c_{ik} of $\mathfrak{A}_{ik}$, then further

$$c_{ik} = e_i c_{ik} = c_{ik} e_k, \quad c_{ik} c_{kl} e_l = c_{il} e_l,$$

hence

$$c_{ik}c_{kl} = c_{il}, \quad c_{ij}c_{kl} = c_{ij}e_j e_k c_{kl} = 0 \quad (j \neq k).$$

The c_{ik} are therefore matrix units (§7).

Construction of K . The assignment

$$a_{ij} = c_{i1}a_{11}c_{1i}$$

assigns to every a_{11} in $\mathfrak{A}_{11}$ a “conjugate element” a_{ii} of $\mathfrak{A}_{ii}$.

This assignment is a ring isomorphism, since the isomorphisms Δ, Γ corresponding to a and c are connected by

$$\Delta_{ii} = \Gamma_{i1}\Delta_{11}\Gamma_{i1}^{-1},$$

a relation well known to represent a ring isomorphism. Now form the elements

$$\alpha = a_{11} + \cdots + a_{nn} = a_{11} + c_{21}a_{11}c_{12} + \cdots + c_{n1}a_{11}c_{1n}.$$

Each a_{11} determines exactly one α (uniquely because the sum is direct). The totality of the α is a field $K \sim \mathfrak{A}_{11}$, since from

$$\alpha = a_{11} + \cdots + a_{nn}, \quad \beta = b_{11} + \cdots + b_{nn},$$

one obtains

$$\alpha + \beta = (a_{11} + b_{11}) + \cdots + (a_{nn} + b_{nn}), \quad \alpha\beta = a_{11}b_{11} + \cdots + a_{nn}b_{nn};$$

so the assignment $a_{11} \mapsto \alpha$ is an isomorphism.

Further,

$$c_{ik}\alpha = c_{ik}a_{kk} = c_{ik}c_{k1}a_{11}c_{1k} = c_{i1}a_{11}c_{1k},$$

$$\alpha c_{ik} = a_{ii}c_{ik} = c_{i1}a_{11}c_{1i}c_{ik} = c_{i1}a_{11}c_{1k},$$

so every α commutes with all c_{ik} .

Finally, the same formulas give

$$c_{ik}K = c_{ik}\mathfrak{A}_{kk} = c_{ik}\mathfrak{r}_k\mathfrak{l}_k = \mathfrak{r}_i\mathfrak{l}_k = \mathfrak{A}_{ik},$$

and therefore

$$\mathfrak{o} = \sum c_{ik}K.$$

The assignment. If every element of $\mathfrak{o}$ is written in the form $\sum c_{ik}\alpha_{ik}$, and the matrix (α_{ik}) is assigned to it, the assignment is an isomorphism, as follows immediately from the definition of matrix multiplication. This proves the main theorem.

Converse. The ring of matrices of degree n over a field A is two-sided simple and completely reducible; A is isomorphic to the automorphism ring of the right ideals, and Ae is the K constructed in the preceding theorem.

Proof. Let the matrix units (§7) be c_{ik} . Put

$$\mathfrak{r}_i = \sum_k c_{ik}A.$$

Then plainly $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$. The $\mathfrak{r}_i$ are simple right ideals, for every non-zero element $\sum c_{ik}\alpha_k$ generates the full $\mathfrak{r}_i$. Indeed, if $\alpha_j \neq 0$, then

$$\left(\sum c_{ik}\alpha_k\right) \cdot (c_{jl}\alpha_j^{-1}) = c_{il},$$

and $\sum c_{il}\beta_l$ runs through all elements of $\mathfrak{r}_i$.

The $\mathfrak{r}_i$ are operator-isomorphic: $\mathfrak{r}_i = c_{ik}\mathfrak{r}_k$.

It follows that $\mathfrak{o}$ is two-sided indecomposable; hence, by the theorem of §13 and the already proved complete reducibility, and since an identity element exists, it is two-sided simple. This is also seen from the fact that any $\sum \sum c_{ik}\alpha_{ik} \neq 0$ generates the entire two-sided ideal $\mathfrak{o}$.

Further,

$$\begin{aligned} \mathfrak{r}_i &= c_{ii}\mathfrak{o}, & \mathfrak{l}_i &= \mathfrak{o}c_{ii}, \\ e &= \sum c_{ii}, & c_{ii} &= e_i, \end{aligned}$$

⁰Recently B. L. van der Waerden found a simpler proof, operating directly with the abstract automorphism field, which is to appear in his book on algebra (Grundlehren d. Math. Wiss., Berlin: Julius Springer). [11 June 1929.]

$$\mathfrak{A}_{ii} = \mathfrak{r}_i \cap \mathfrak{l}_i = c_{ii}A \sim A.$$

Finally, putting $a_{11} = c_{11}\lambda$, one obtains

$$\alpha = a_{11} + c_{21}a_{11}c_{12} + \cdots = c_{11}\lambda + c_{22}\lambda + \cdots = e\lambda,$$

which proves everything asserted.

The center of $\mathfrak{o}$ is the center of K , hence a field.

Proof. $\mathfrak{Z}(K)$ consists of all ζ that commute with all x . But these also commute with all c_{ik} , and hence with all sums $\sum c_{ik}a_{ik}$. Thus

$$\mathfrak{Z}(K) \subseteq \mathfrak{Z}(\mathfrak{o}).$$

Conversely, let z lie in $\mathfrak{Z}(\mathfrak{o})$:

$$z = \sum_{\mu} \sum_{\nu} c_{\mu\nu} \gamma_{\mu\nu}.$$

Then

$$zc_{ik} = c_{ik}z, \quad \sum_{\mu} c_{\mu k} \gamma_{\mu i} = \sum_{\nu} c_{i\nu} \gamma_{k\nu},$$

and so

$$\begin{aligned} \gamma_{\mu i} &= 0 \quad (\mu \neq i), & c_{ik} \gamma_{ii} &= c_{ik} \gamma_{kk}, & \gamma_{ii} &= \gamma_{kk} = \gamma; \\ z &= \sum c_{ii} \gamma = e\gamma = \gamma. \end{aligned}$$

Thus z lies in K and commutes with all x , hence z lies in $\mathfrak{Z}(K)$.

Since a matrix ring is of course also left completely reducible, and since every right completely reducible ring with identity element is a direct sum of matrix rings, it follows that:

Every right completely reducible ring with identity element is also left completely reducible, and conversely.

And: The center of a completely reducible ring with identity element is a direct sum of commutative fields corresponding to the two-sided simple matrix rings.

There remains the following theorem:

Any two different decompositions $\mathfrak{o} = \sum c_{ik}K$, $\mathfrak{o} = \sum c'_{ik}K'$ are transformed into one another by inner automorphisms $\alpha' \mapsto x^{-1}\alpha'x$, $c'_{ik} \mapsto x^{-1}c_{ik}x$.

Proof. Let

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = \mathfrak{r}'_1 + \cdots + \mathfrak{r}'_n,$$

and let a, b be elements mediating two reciprocal isomorphisms of the ideals $\mathfrak{r}_1, \mathfrak{r}'_1$:

$$\mathfrak{r}_1 = a\mathfrak{r}'_1, \quad \mathfrak{r}'_1 = b\mathfrak{r}_1, \quad ab = c_{11}, \quad ba = c'_{11}.$$

Put

$$x = \sum_i c_{i1} a c'_{1i}, \quad y = \sum_i c'_{i1} b c_{1i}.$$

Then

$$xy = e, \quad y = x^{-1}, \quad x^{-1}c_{ik}x = c'_{ik}, \quad x^{-1}\alpha x = \alpha'.$$